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Original Article

Charge matters: how flanking substrate charge modulates O-glycan Core elongation

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Mucin type O-glycan core elongation is typically performed by the C1GALT1, B3GNT6, and ST6GalNAc-I/-II O-glycosyltransferases. These enzymes target the Tn antigen (GalNAc-O-Thr/Ser) dictating the fate of O-glycan elongation, playing important roles in health and disease. Changes in transferase expression and glycan structure are commonly associated with diseases such as cancer, Tn-syndrome, and ulcerative colitis. Despite their significance, their substrate specificities and their biological roles remain elusive. Here, we examine the roles of flanking glycopeptide substrate charge using a library of differently charged glycopeptides and a small library of PSGL-1 Thr57 based charged glycopeptides. We found that C1GALT1 was most influenced by flanking charge preferring negatively charged substrates, while B3GNT6 and ST6GalNAc-II were less influenced, showing unique N- and C-terminal charge preferences. Interestingly, ST6GalNAc-I was not influenced by flanking charge. These charge specificities were further maintained against the charged PSGL-1 glycopeptides, although ST6GalNAc-I showed an increased preference towards a remote N-terminal positive charge. The observed charge preferences were to a large part driven by substrate interactions with the electrostatic surface of the transferase. We propose that negative flanking charge may assist C1GALT1 in targeting key glycosites such as in PSGL-1 and podoplanin. Our findings are consistent with a Golgi hierarchy, where the cis-Golgi localized GalNAc-Ts and C1GALT1 determine the site and thus fate of glycosylation, while the trans-Golgi less-specific ST6GalNAc-I provides a final capping function. This characterization of substrate charge preference furthers our understanding of how these enzymes select their substrates and may contribute to our understanding of their biological roles.

Keywords: Core glycosyltransferases/electrostatic interactions/GalNAc-T/Golgi/O-glycan elongation.

Introduction

Mucin type O-glycosylation is a diverse, ubiquitous, and essential post-translational modification of proteins that is initiated by a family of 20 UDP-N-acetyl-D-galactosamine: polypeptide N-acetylgalactosaminyltransferases (GalNAc-Ts). These transferases catalyze the transfer of α-GalNAc from UDP-GalNAc donors to the hydroxyl groups of Ser/Thr residues of peptide and protein substrates, forming the Tn antigen (GalNAc-α-O-Ser/Thr). How the GalNAc-Ts choose their substrates has been extensively studied; revealing unique but overlapping peptide sequence specificities, varying preferences for Thr over Ser acceptors, and isoform specific recognition of neighboring and remote prior GalNAc glycosylation (Fritz et al. 2006; Gerken et al. 2006; Wandall et al. 2007; Gerken et al. 2008; Raman et al. 2008; Gerken et al. 2011, 2013; Bennett et al. 2012; Revoredo et al. 2016; de las Rivas et al. 2018, 2019, 2017; Daniel et al. 2020). Recently, the profound effects of flanking charged residues was reported revealing further diversity in their specificities (Ballard et al. 2023). From these features, and especially from the different glycopeptide specificities, the initiation of O-glycosylation is likely an ordered process in most cells.

Once formed, the GalNAc-α-O-Ser/Thr is typically elongated by one or more Golgi resident glycosyltransferases including the C1GALT1 (Core-1 transferase or T-synthase) forming the core-1 structure (T antigen: Galβ1–3GalNAc-α-O-Ser/Thr) and the B3GNT6 (Core-3 transferase) forming the core-3 structure (GlcNAcβ1–3GalNAc-α-O-Ser/Thr). The T antigen and the core 3 structure can be further elongated by a variety of glycosyltransferases to form a series of diverse O-glycan structures (Fig. 1) (Bergstrom et al. 2017; Xia et al. 2004; Robbe et al. 2004). Subsequentially, both the ST6GalNAc-I and -II transferases can synthesize the sialyl Tn (STn) antigen (Neu5Acα2–6GalNAc-α-O-Ser/Thr) and the Sialyl-6 T antigen (Galβ1–3 (Neu5Acα2–6) GalNAc-α-O-Ser/Thr) although with different preferences (see below) (Fig. 1). Additionally, the ST6GalNAc's can sialylate the GalNAc on the core 3 structure to terminate further glycan elongation (Marcos et al. 2004; Sewell et al. 2006). The elongation of the Tn antigen by the above transferases is biologically significant since the Tn antigen is not normally detected except during development and other diseases including cancers (Akita et al. 2001; Ju et al. 2011; Ju et al. 2013; Ju et al. 2014). Despite our extensive knowledge on

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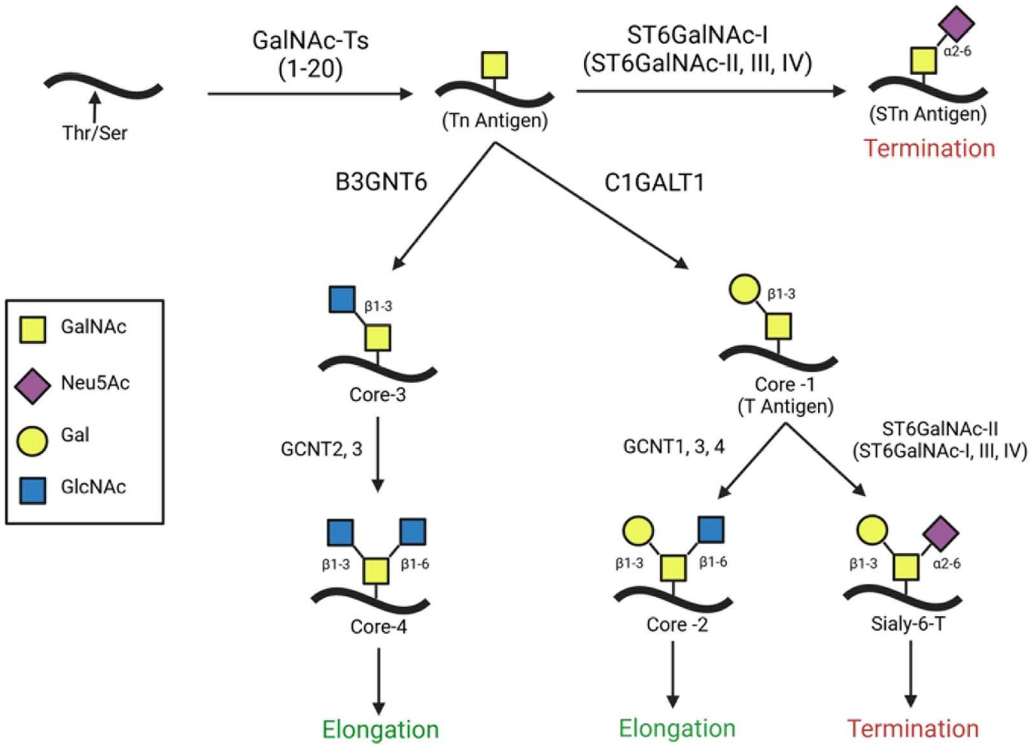


Fig. 1. Major O-glycan initiation and elongation pathways. Mucin type O-glycosylation is initiated by a family of Golgi resident GalNAc-Ts forming the Tn antigen (GalNAc-α-O-Ser/Thr). The Tn antigen is usually further elongated by either B3GNT6 or C1GALT1 to form the core-3 (GlcNAcβ1–3GalNAc-α-O-Ser/Thr) or core-1 (Galβ1–3GalNAc-α-O-Ser/Thr: T antigen) glycan structures, respectively or sialylated by the ST6GalNAc-I (-II, -III, or IV) forming the STn antigen (Neu5Acα2–6GalNAc-α-O-Ser/Thr) thus terminating further glycan elongation. The core-3 structure can be further elongated by GCNT-2 and -3 forming the core-4 glycan structure allowing for further elongation into more complex glycan structures. Similarly, the T antigen can be further elongated by GCNT-1, −3, and − 4 to form the core-2 structure that can be subsequently elongated. Alternatively, the T-antigen can be sialylated by primarily the ST6GalNAc-II (-I, -III, or IV) forming the Sialy-6-T (Galβ1–3 (Neu5Acα2–6) GalNAc-α-O-Ser/Thr) structure which terminates further elongation. Note that there are additional, less prevalent core structures and other elongated structures not shown (Moremen et al. 2012; Cervoni et al. 2020; Brockhausen et al. 2022; Chrysinas et al. 2024). (figure created in BioRender. https://BioRender.com/f54e887.)

how GalNAc-Ts choose their targets, our understanding on how the core O-glycan elongating transferases target their glycopeptide substrates remains limited. Currently, the actual biologically important and disease associated target sites of these core transferases are largely unknown, therefore, elucidating their specificities will assist in understanding their roles in health and disease.

The C1GALT1 transferase is ubiquitously expressed across all tissues and plays critical roles in fly and mouse development, cancer progression, and disorders such as Tn syndrome, IgA nephropathy, colitis, and gastritis (Xia et al. 2004; Ju and Cummings 2005; Brockhausen 2006; Suzuki et al. 2008; Yoshida et al. 2008; Cazet et al. 2010; Ju et al. 2013; Chou et al. 2015; Bergstrom et al. 2016, 2017; Gao et al. 2016; Bergstrom et al. 2017; Liu et al. 2020; Sun et al. 2021; Wan and Yu 2021; Xia et al. 2022; Erger et al. 2023). What governs the overall specificity of the C1GALT1, and the identities of its targeted glycosylation sites are not well known. Studies against libraries of short glycopeptide substrates have shown that the mammalian C1GALT1s are influenced by glycopeptide sequence, length, structure, and glycosylation density, however, it is not obvious what C1GALT1 recognizes as a fundamental glycopeptide motif (Ju et al. 2002; Brockhausen et al. 1992; Granovsky et al. 1994; Brockhausen et al. 1990; Gao et al. 2013). Interestingly, studies have also shown that the presence of neighboring core 1 glycans and GalNAc residues decrease the activity of C1GALT1 towards glycopeptides and the porcine salivary gland mucin (Granovsky et al. 1994; Gerken 2004; Brockhausen et al. 2009). A broader random glycopeptide study (Perrine et al. 2009) showed that *Hs*C1GALT1 has slightly increased selectivity towards negatively charged substrates supporting previous observations that negative charged substrates increases C1GALT1 activity (Brockhausen et al. 1990; Granovsky et al. 1994; Brockhausen et al. 2009). This study also showed increased preferences for Gly at the +1 and Phe/Tyr at the +3 positions relative to the acceptor GalNAc-α-O-Thr (Perrine et al. 2009). Only recently has the crystal structure of the *Drosophila melanogaster* (*Dm*) C1GALT1 been solved having ~43% identity with the human homologue providing information on the mechanistic nature driving GalNAc moiety recognition. Surprisingly, in this structure, the acceptor GalNAc-α-O-Thr of the bound glycopeptide is oriented in an energetically unfavored conformation, suggesting that C1GALT1 has a unique GalNAc-α-O-Thr/Ser conformational preference (González-Ramírez et al. 2022). In this work it was also suggested that the *Dm*C1GALT1 possessed similar glycopeptide specificities as that observed in the random glycopeptide studies of *Hs*C1GALT1 (Perrine et al. 2009).

B3GNT6 is primarily expressed in normal gastrointestinal tissues (Vavasseur et al. 1994; Iwai et al. 2005) and when lost in the mouse results in increased susceptibility to colitis and colorectal carcinomas (An et al. 2007; Bergstrom et al. 2017). B3GNT6 expression is down regulated in colorectal,

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gastric, and pancreatic cancers which together with studies in intestinal cell lines suggests B3GNT6 may act as a tumor suppressor (Vavasseur et al. 1994; Iwai et al. 2005; Radhakrishnan et al. 2013). Although studies against a series of glycopeptide substrates have been performed, no clear understanding of its specificity has been revealed except that it differs from C1GALT1 and that modifications to the GalNAc moiety impact activity (Gao et al. 2013; Brockhausen et al. 1990; Vavasseur et al. 1995; Iwai et al. 2002).

The STn antigen is typically synthesized by the ST6GalNAc-I or ST6GalNAc-II, however, in the normal cellular context the ST6GalNAc-I dominates (Harduin-Lepers et al. 2001; Marcos et al. 2011; Munkley 2016). ST6GalNAc-I is most active against the Tn antigen (Ikehara et al. 1999; Marcos et al. 2004) while ST6GalNAc-II is most active against the T antigen and forms the Sialyl-6 T antigen (Galβ1–3 (Neu5Acα2–6) GalNAc-α-O-Ser/Thr) (Samyn-Petit et al. 2000; Marcos et al. 2004). The presence of the STn antigen is commonly associated with the loss of expression of C1GALT1 found in many cancers (Munkley 2016), and is correlated with increased cell invasiveness and chemoresistance (Barrow et al. 2013; Ogawa et al. 2017; Wang et al. 2019). Interestingly, ST6GalNAc-I and -II play varying roles in different cancers such as colorectal and breast either promoting or impeding cancer progression (Barrow et al. 2013; Murugaesu et al. 2014; Miao and Zhao 2016; Liu et al. 2017; Ogawa et al. 2017; Luo et al. 2023) Thus, the varying physiological impacts these transferases have in health and disease may be due to their differences in specificities toward their acceptor glycan structures. Although both ST6GalNAc-Ts show efficient sialylation towards asialofetuin, asialo-bovine submaxillary mucin, and asialo-ovine submaxillary mucin (Marcos et al. 2004; Tsuji and Takashima 2014) their specificity has not been characterized with respect to glycopeptide substrate sequence.

We recently showed that the GalNAc-Ts utilize their electrostatic surface potentials to interact with charged residues remote to the acceptor site to profoundly modulate glycosylation (Ballard et al. 2023). Based on these studies, it seemed reasonable that flanking substrate charge and transferase surface electrostatics might be a factor modulating core transferase specificity. To examine how flanking substrate charge impacts the core transferases we obtained the GalNAc-α-O-Thr versions of the charged peptide substrates previously used for the GalNAc-Ts (Ballard et al. 2023) (Table 1). These substrates were semi-synthetically produced by fully GalNAc glycosylating the "naked" peptides with GalNAc-T1 and subsequentially glycosylating with the core transferases. This approach was used to characterize *Dm*C1GALT1, *Hs*ST6GalNAc-I and -II, and their GFP versions (GFP-*Dm*C1GALT1, GFP-*Hs*ST6GalNAc-I, and GFP-*Hs*ST6GalNAc-II). In addition, the GFP version of *Hs*B3GNT6, GFP-*Hs*B3GNT6, was also characterized. We found that the core elongating transferases were variably impacted by flanking charge, with *Dm*C1GALT1 (Non-GFP and GFP tagged) showing significantly increased activity towards negatively charged substrates compared to neutral and positive charged substrates. In contrast, the GFP-*Hs*B3GNT6 and *Hs*ST6GalNAc-II (Non-GFP and GFP) showed weak N- and C-terminal charge specificities while the *Hs*ST6GalNAc-I (Non-GFP and GFP) showed no significant charge preferences towards these substrates. These results are consistent with the electrostatic surface potentials of C1GALT1, B3GNT6, and ST6GalNAc-II while at odds to that of ST6GalNAc-I. Nevertheless, these findings suggest for the most part flanking charge may be an important factor modulating core elongation. This was confirmed by characterizing the glycosylation of a series of differently charged glycopeptides derived from the biologically important Thr57 glycosylation site of the mucin like P-selectin glycoprotein ligand-1 (PSGL-1). Here flanking charge affects were observed for all the transferases studies including ST6GalNAc-I. In summary, our results show that the specificity of *Dm*C1GALT1 is significantly influenced by the nature of the peptide charge while the other core transferases are less so affected. Together, this may further suggest that C1GALT1 may act to control the site-specific fate of O-glycan elongation (Ju and Cummings 2002; Ju et al. 2008; Barrow et al. 2013; Steentoft et al. 2013; Sun et al. 2021; Aryal et al. 2024).

Results

The activities of the core elongating transferases are variably impacted by flanking substrate charge

To determine the substrate charge specificities of the core elongating transferases we characterized their activity against the 10 semi-synthetically produced model glycopeptide substrates in Table 1 as described in the Methods. These model substrates were designed to contain all possible combinations of N- and C-terminal charge. Positive charge represented by a triad of Arg residues (abbreviated R), negative charge by a triad of aspartic acid residues (abbreviate D), and neutral charge by either Gly-Ala-Gly (abbreviated G) or Gln-Ala-Gly (abbreviated Q). These residues were placed flanking the glycosylated GalNAc-T3 -YAVTPGP- optimal motif as described in Ballard et al. 2023. The extent of relative glycopeptide glycosylation for the 10 glycopeptides and a blank control are shown in the left panels of Figs 2 and 4–6 from the most positive to most negative substrate for each transferase studied. The right panels show the same data but rearranged with a fixed N- or C-terminal charge providing an additional visual showing the effects of varying charge. Below the plots we show the transferase's surface electrostatic potentials modeled with their preferred charged glycopeptide substrate for comparison to the obtained charged preferences.

DmC1GALT1

WT-*Dm*C1GALT1 shows highest activity toward negatively charged glycopeptide substrates with the highest glycosylation observed for the DG and DD substrates (Fig. 2A and B and Supplementary Fig. S2). WT-*Dm*C1GALT1 is a dimer in solution stabilized by an N-terminus loop and an unstructured C-terminus loop (González-Ramírez et al. 2022) (Fig. 2E). The electrostatic surface potential of the WT-*Dm*C1GALT1 dimer shows moderately dense regions of positive charge surrounding the known substrate binding site that likely enhance the binding of negatively charged substrates through direct electrostatic interactions, similar to what was observed for a number of the GalNAc-Ts (Ballard et al. 2023). Results for the monomeric GFP-*Dm*C1GALT1 are nearly identical to the WT-*Dm*C1GALT1 (Figs. 2C and D). Thus, the disruption of the dimer and the presence of the highly charged GFP tag has no significant impact on charge preferences compared to the WT-*Dm*C1GALT1 dimer against these substrates,

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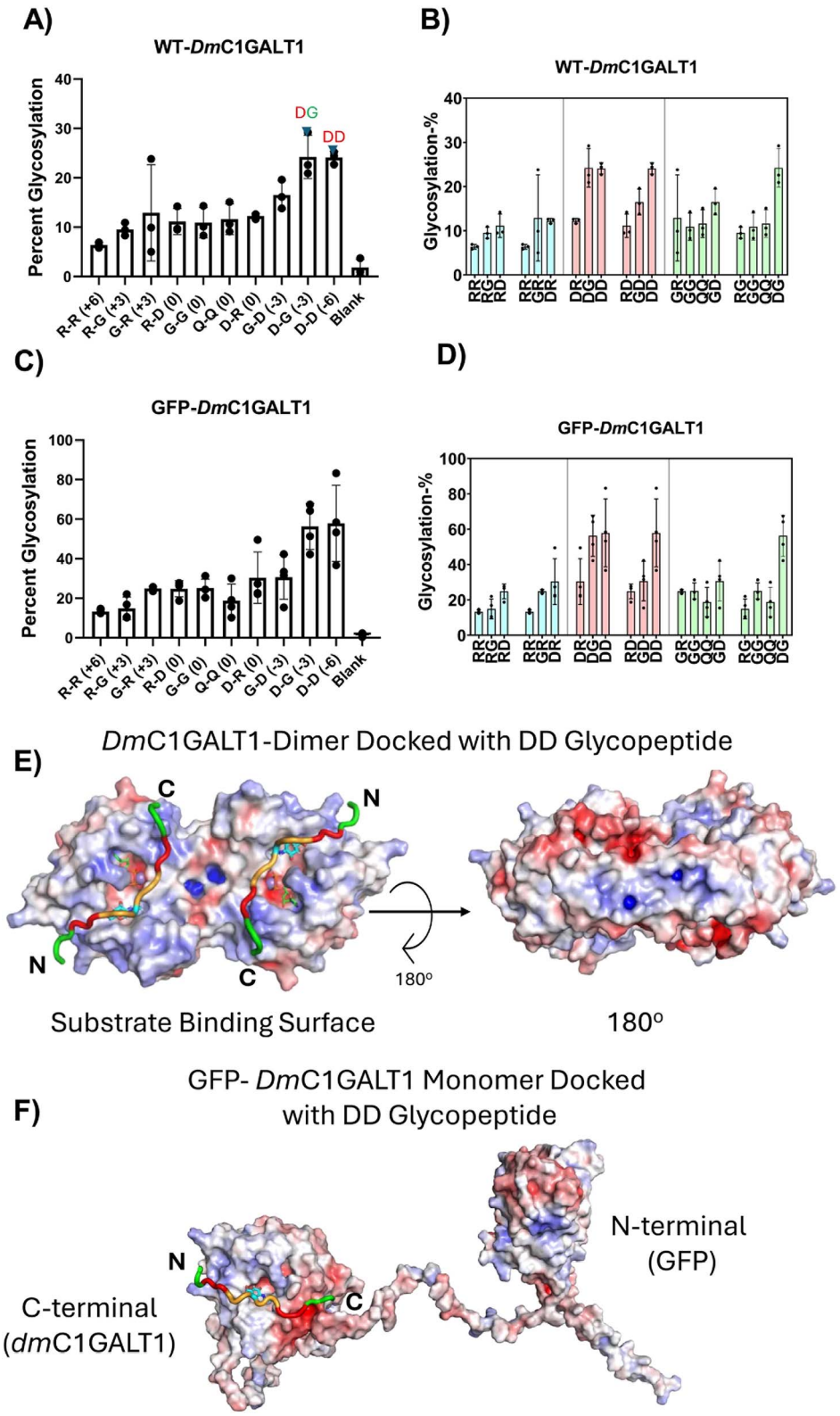


Fig. 2. *Dm*C1GALT1 prefers negatively charged Glycopeptide substrates. Glycosylation of the charged glycopeptides (0.47 mM) in Table 1 by WT-*Dm*C1GALT1 (0.027 μM) and GFP-*Dm*C1GALT1 (0.027 μM) (**A**–**D** respectively). In **A** and **C**) substrates are arranged with the most positive RR glycopeptide on the left and the most negative DD glycopeptide is on the right. In **B**) and **D**), the data is rearranged to evaluate the effects of altering N- or C-terminal charge while the opposing N- or C-terminal charge is maintained. Blue represents a constant positive N- or C-terminal charge (R), red represents a constant negative N- or C- terminal charge (D), and green represents a neutral N- or C-terminal charge (G). **E**) the front and back views of the electrostatic surface charge distribution of the dimeric WT-*Dm*C1GALT1 modeled with bound DD glycopeptide substrate (PDB: 7Q4I). **F**) the electrostatic surface charge distribution of the monomeric GFP-*Dm*C1GALT1 modeled with a bound DD glycopeptide substrate. Note that the GFP is tethered to the N-terminus of monomeric *Dm*C1GALT1 in the figure. In **E** and **F**, the glycopeptide colors are labeled as follows: Green for the GAGA N- and C-terminal residues, red for the triad of negatively charge asp residues, orange for the +/− 3 -YAV/PGP- residues flanking GalNAc-α-O-Thr, and cyan for the GalNAc-α-O-Thr acceptor. The N- and C- terminal ends of the bound glycopeptides are labeled N and C respectively. See Supplementary Fig. S3 for statistical analysis of the data in A and C. Experiments were repeated three to four times.

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Table 1. Charged glycopeptide substrates.

GAGAXXXYAVT∗PGPXXXAGAG Substrate ID and Net Charge	where XXX = RRR (R), DDD (D), or GAG (G) Substrate Sequence
RR (+6)	GAGARRRYAVT∗PGPRRRAGAG
RG (+3)	GAGARRRYAVT∗PGPGAGAGAG
GR (+3)	GAGAGAGYAVT∗PGPRRRAGAG
RD (0)	GAGARRRYAVT∗PGPDDDAGAG
GG (0)	GAGAGAGYAVT∗PGPGAGAGAG
QQ (0)	GAGAQAGYAVT∗PGPGAQAGAG
DR (0)	GAGADDDYAVT∗PGPRRRAGAG
GD (−3)	GAGAGAGYAVT∗PGPDDDAGAG
DG (−3)	GAGADDDYAVT∗PGPGAGAGAG
DD (−6)	GAGADDDYAVT∗PGPDDDAGAG

although the GFP-*Dm*C1GALT1 shows a C-terminal negative charge patch (Fig. 2F). In addition, as GFP is highly polarized it might be expected to alter substrate charge preferences (Supplementary Fig. S4), however, the model of GFP-*Dm*C1GALT1 reveals the GFP to be tethered at a distance from the *Dm*C1GALT1 suggesting its presence would have minimal effects (Fig. 2F). We also examined the surface electrostatics of the modeled monomeric and dimeric human (*Hs)* C1GALT1 for comparison to the *Dm*C1GALT1 used in this study (Supplementary Fig. S3). The monomeric *Hs*C1GALT1 showed similar surface electrostatic distributions to that of the monomeric *Dm*C1GALT1, however, substrate binding surface of the dimeric model of *Hs*C1GALT1 is somewhat more negative. This suggests the *Hs*C1GALT1 may have slightly weaker negative charge substrate preferences compared to that observed for the *Dm*C1GALT1. However, we presently do not know how the disordered regions of *Hs*C1GALT1 may fold and how this may affect its charge preferences.

To confirm that charge–charge interactions are responsible for *Dm*C1GALT1's charge specificity, studies at elevated ionic strength (400 mM) that would partially disrupt charge–charge interactions between the enzyme and the substrate were performed (Fig. 3A and B and Supplementary Fig. S5A). Under these conditions, the percent glycosylation of the least preferred positive substrates (i.e. RR, RG, and GR) increased 1.5 to 2-fold while the preferred negative charge substrates (GD, DG, and DD) and the neutral uncharged substrates (GG and QQ) remained relatively unchanged (Fig. 3C). This was expected for the positive charged and neutral uncharged glycopeptide substrates but unexpected for the negative charged glycopeptide substrates which would have been expected to decrease in activity under higher ionic strength conditions. We, therefore, performed experiments at higher ionic strength (700 mM) which revealed relatively uniform activity across our series of charged glycopeptide substrates (Fig. 3D and E and Supplementary Fig. S5B). Under these conditions, the expected decreased activity towards the most negatively charged DD substrate is observed while the other negatively charged substrates (GD and DG) remained relatively unchanged (Fig. 3E and F). Additionally, the activities towards the most positive charges substrates (RR, RG, and GR) further increased to 2–2.6-fold compared to 1.5–2-fold for the lower (400 mM) ionic strength studies (Fig. 3C and F). Together, these data suggest that under the lower (∼100 mM) more physiological conditions the positive surface charge of *Dm*C1GALT1 repels the positive charge substrates while strongly attracting the negative charged substrates. Thus, for both monomeric and dimeric *Dm*C1GALT1, enzyme-substrate charge–charge interactions are significantly contributing to the transferase's overall negative charge preference.

B3GNT6

GFP-*Hs*B3GNT6's charge specificities are more subtle than for C1GALT1 with the neutral DR glycopeptide having the highest activity and the GD glycopeptide the least by 4-fold (Figs. 4A and B). (Note, the GFP version of B3GNT6 was characterized due to the poor stability of WT-B3GNT6). Interestingly, the N-terminal Arg glycopeptides (RR, RG, and RD) have activities significantly lower than the DR glycopeptide while the GR, GG, QQ, DG, and DD glycopeptides show even lower activities with the GD and DD having the least activity (Fig. 4A and Supplementary Fig. S6). As there is no obvious charge-activity relationship patterns against these substrates with *Hs*B3GNT6, it is likely that each substrate may bind to different charged surfaces on the enzyme similar to what we have observed for a subset of the GalNAc-Ts (Ballard et al. 2023). To date there are no crystal structures of *Hs*B3GNT6, therefore, we used Alphafold 3 (Abramson et al. 2024) to model the GFP-*Hs*B3GNT6 (Fig. 4C and D). The model reveals an expected high negative charge around the predicted Mn^{2+} binding site surrounded by patches of different charged surfaces of varying strength and size (Fig. 4C and D). In addition, a large positive region is observed on the reverse side of the enzyme. The patchwork of charged around *Hs*B3GNT6's substrate binding site are, therefore, likely responsible for its observed unique charge specificity.

ST6GalNAc-II

WT-*Hs*ST6GalNAc-II, is most active against the DR, followed by the GR and DG glycopeptides while having the least activity against the RD and GD glycopeptides, suggesting preference for negative N-terminal and positive C-terminal charged substrates (Figs. 5A and B). GFP-*Hs*ST6GalNAc-II shows similar, but weaker, preferences as the wild type transferase (Figs. 5C and D). The electrostatic surface potential of the substrate binding site (CMP-Neu5Ac binding) based on its x-ray structure (Moremen et al. 2018) is roughly neutral with adjacent pockets of negative charge, while the surrounding surfaces shows mostly weak positive charge (left structure of Fig. 5E and F). Thus, our observed weak N- and C-terminal charge preferences appear consistent with *Hs*ST6GalNAc-II's

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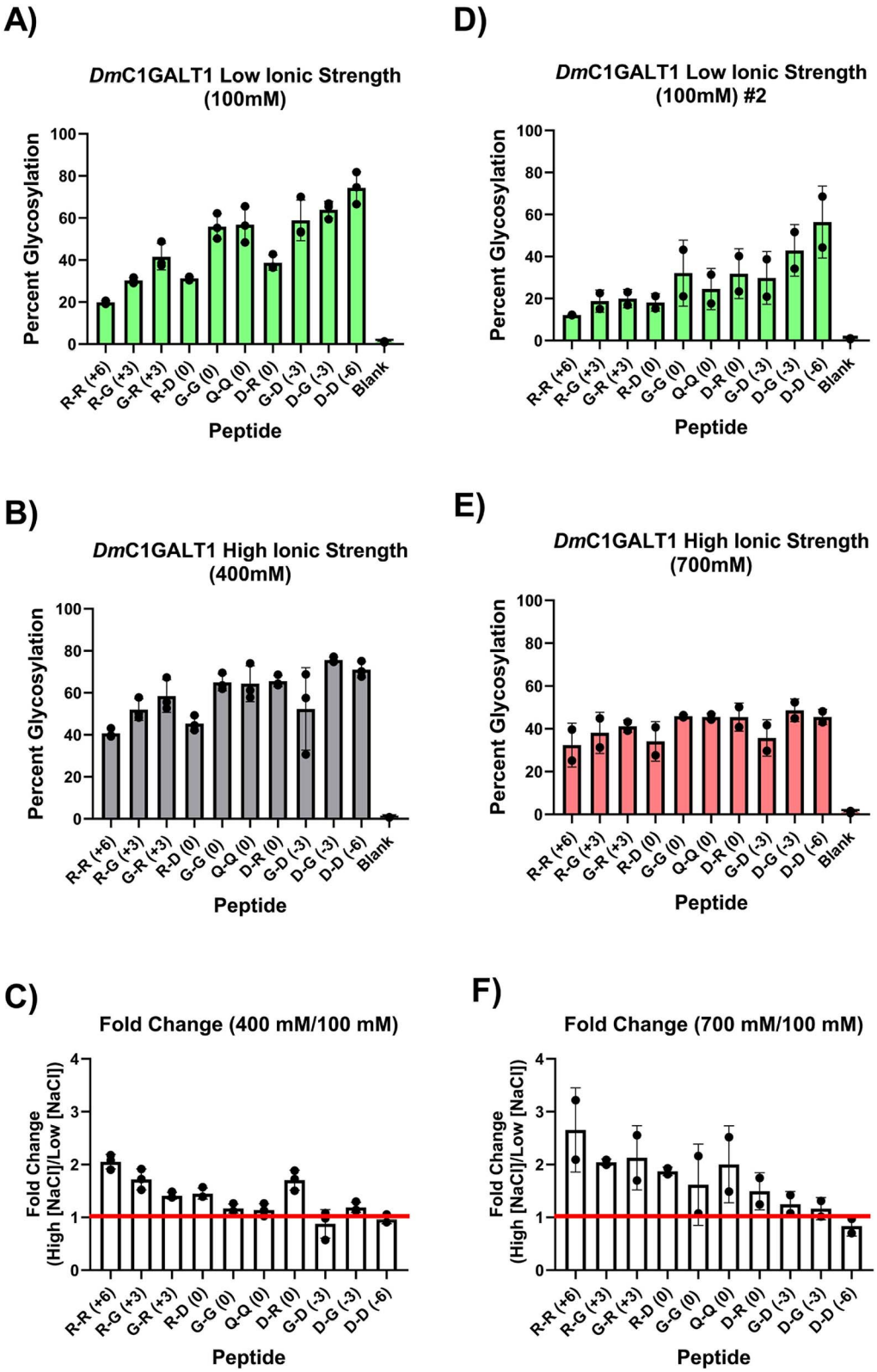


Fig. 3. Elevated ionic strength disrupts WT-*Dm*C1GALT1 charge–charge substrate-enzyme interactions. Glycosylation of the charged glycopeptides (0.47 mM) by WT-*Dm*C1GALT1 (0.027 μM) under normal (**A** and **D**) and elevated (**B** and **E**) ionic strength. Plots are arranged as described in Fig. 2. **A** and **B**, normal (∼100 mM) and 400 mM ionic strength reaction conditions, **D** and **E** comparison of normal (∼100 mM) and 700 mM ionic strength reactions. **E** and **F**, plots of the fold change of activity (high/low ionic strength) for the 400 and 700 mM ionic strength comparisons. See Supplementary Fig. S5 for statistical analysis. Experiments were repeated two to three times.

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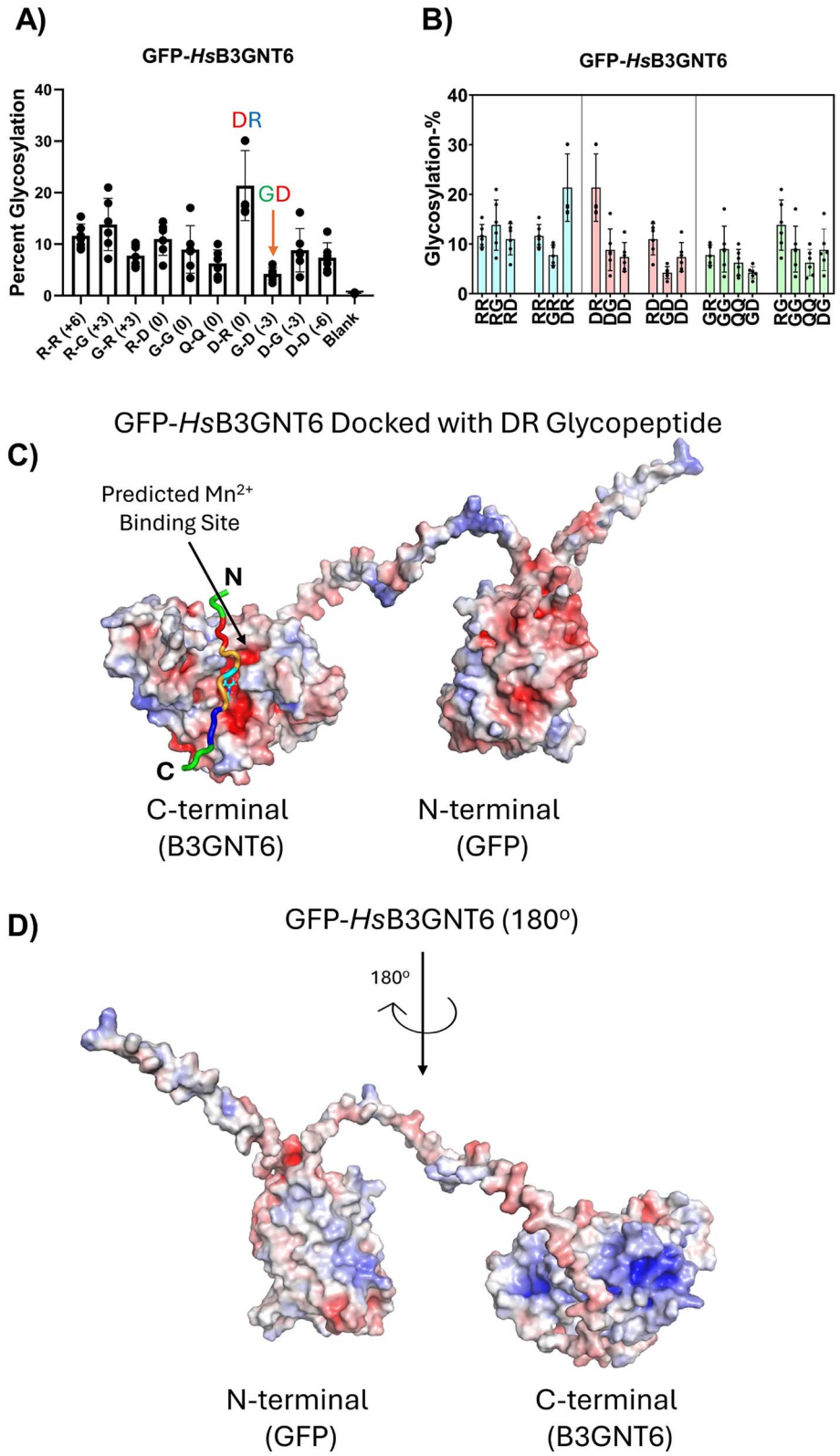


Fig. 4. Flanking substrate charge variable impacts the specificity of *Hs*B3GNT6. Glycosylation of the charged glycopeptides in Table 1 by GFP-*Hs*B3GNT6 (1.2 μM), panels **A** and **B**) (arranged identically as described in Fig. 2A and B). Glycopeptides with the highest and lowest activities are labeled and color coded to represent their flanking N- and C-terminal charge. **C** and **D**) the front and back views of the electrostatic surface charge of modeled GFP-B3GNT6 with the DR peptide bound. Colors of the peptide are identical to that given in Fig. 2E-F where blue represents the triad of positive charged C-terminal Arg residues. See supplementary fig. 6 for statistical analysis. Experiments were repeated six times.

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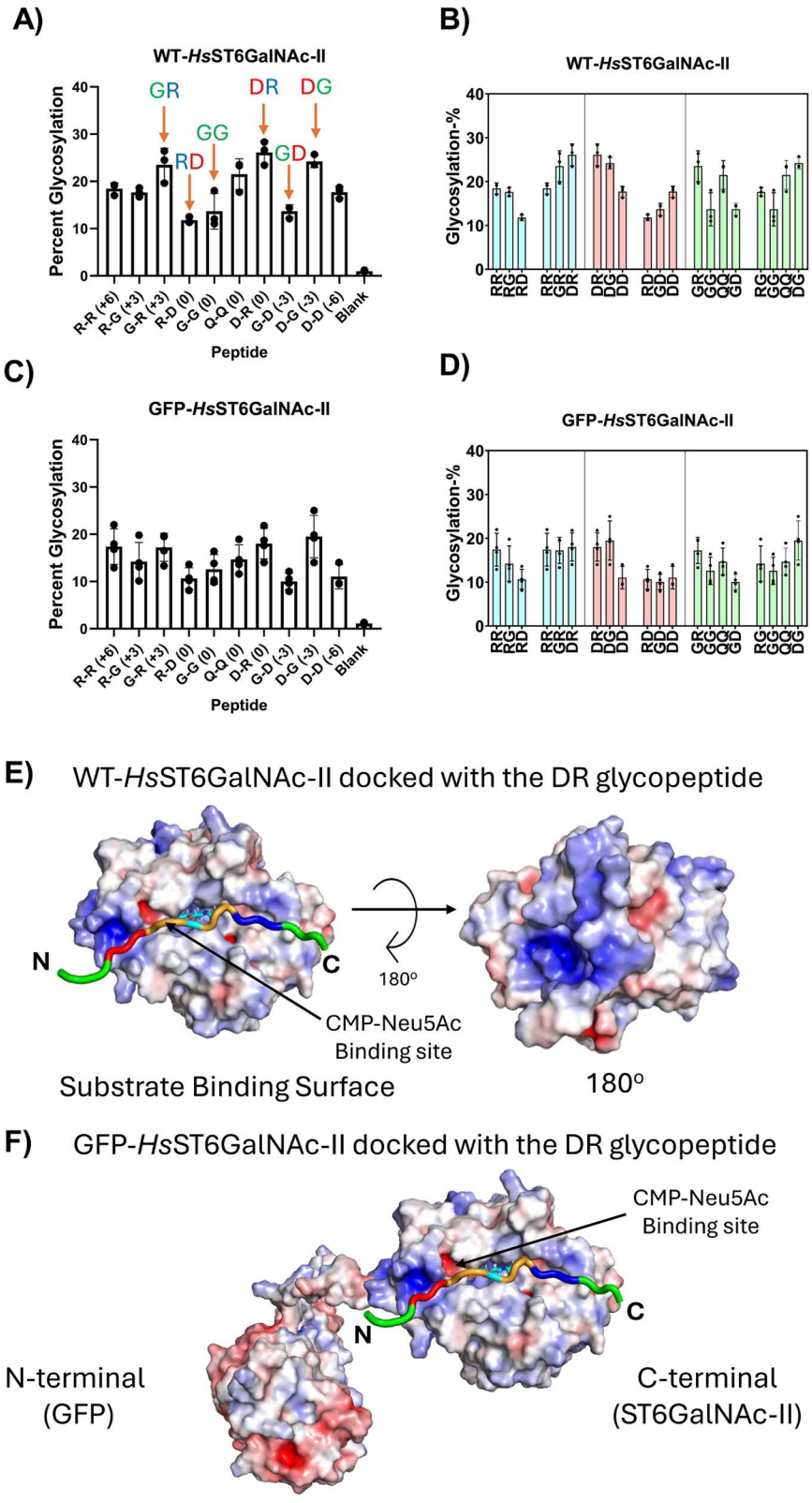


Fig. 5. *Hs*ST6GalNAc-II prefers negative N- and positive C-terminal substrates. Glycosylation of the charged glycopeptides (0.47 mM) in Table 1 by WT-*Hs*ST6GalNAc-II (0.02 μM) and GFP-*Hs*ST6GalNAc-II (0.02 μM), panels A-D (arranged as in Fig. 2A–D). Glycopeptide substrates having the highest percent glycosylations (GR, DR and DG) and lowest (RD, GG, and GD) are labeled and color coded. **E** and **F**) electrostatic surface charge of WT- (PDB: 6APL) and GFP-ST6GalNAc-II with the DR glycopeptide bound are arranged and color coded as described in Figs. 2 and 4. See Supplementary Fig. S7 for statistical analysis. Experiments were repeated three times.

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surface charge. As with C1GALT1 the tethered GFP tag shows no significant effect on *Hs*ST6GalNAc-II's charge preferences.

ST6GalNAc-I

In contrast to the previous transferases, WT- and GFP-*Hs*ST6GalNAc-I show nearly uniform activity for all 10 substrates indicating no specific preferences towards glycopeptide charge using our glycopeptide substrates (Fig. 6A–D). Interestingly, the homology modeled electrostatic surface potential of *Hs*ST6GalNAc-I shows many areas of relatively high positive charge (left structure of Fig. 6E and F) significantly greater than those observed for ST6GalNAc-II (Fig. 5E). However, these positive charged patches apparently do not modulate the binding of the different charged glycopeptides. Based on these results it appears that *Hs*ST6GalNAc-I is unaffected by the flanking charge of these substrates, thus its activity appears to be mainly driven by the presence of the Thr-O-GalNAc. The differences between the charge specificities of *Hs*ST6GalNAc-I and -II may be due to their reported specificities towards the Tn and T antigen, respectively (Samyn-Petit et al. 2000; Marcos et al. 2004; Marcos et al. 2011; Munkley 2016). This may cause a difference in how the Thr-O-GalNAc acceptor is presented in the active site therefore impacting their overall specificities towards differently charged substrates. Additionally, the modeled catalytic domain of *Hs*ST6GalNAc-I (~24.5 kDa) is significantly smaller than that of *Hs*ST6GalNAc-II (34.9 kDa) possibly impacting how these transferases interact with underlying peptide sequences (Supplementary Fig. S15). However, as we show below, *Hs*ST6GalNAc-I may recognize more remote charged residues in the positively charged PSGL-1 glycopeptide substrates.

Potential role of flanking charge for targeting site specific O-glycosylation using PSGL-1 as a model

P-selectin glycoprotein ligand-1 (PSGL-1) is an extended cell surface heavily O-glycosylated mucin expressed on most leukocytes and activated endothelial cells. Its weak binding to the selectins on endothelial cells initiates the rolling of circulating leukocytes as the first step in the recruitment and extravasation of leukocytes at sites of inflammation (Li et al. 1996). PSGL-1 binding to its selectin receptors requires both an extended Core 2 sialyl-Lewis X determinant (SLe^x) (Fig. 7A) at Thr57 as well as specific sulfation of Tyr residues N-terminal of this site (Liu et al. 1998; Xia et al. 2003; Goth et al. 2023). It appears that only Thr57 out of the ~71 other Ser and Thr glyco-sites contains this structure suggesting a targeted biosynthesis of this site. How the Thr57 site is specifically targeted for SLe^x elongation is not well understood. However, since this site has the highest negative charge density of PSGL-1 (Supplementary Fig. S9) we previously suggested (Gerken et al. 2011) that this high negative charge might potentially serve as one aspect of its targeting by greatly enhancing the formation of the Tn structure by a subset of GalNAc-Ts, followed by rapid glycosylation by C1GALT1 forming the core-1 structure (Fig. 1). Thus, if initiation at this site is more rapid than all other sites, then kinetically it would be more likely to be further elongated to the sialyl-Lewis X structure. Our original random peptide preferences for a subset GalNAc-Ts and C1GALT1 indeed were consistent with this hypothesis where we suggest GalNAc-T1 may be the primary initiating GalNAc-T glycosylation of Thr57. Although this had not been experimentally shown, it is consistent with mouse knockout studies of GalNAc-T1 (Tenno et al. 2007). Therefore, to address this hypothesis we obtained a series of differently charged peptides and glycopeptides containing the *Hs*PSGL-1 Thr57 sequence, as shown in Fig. 7B. These were named RR, Rn, NN, nR, and nn (WT) to represent the N- or C-terminal substitutions where n represents the native sequence and R and N to represent positive charged (Arg) or neutral charged (Gln or Asn) residue substitutions, respectively. (Note, the GFP-B3GNT6 was excluded from these studies as it was inactive when these experiments were conducted.)

Glycosylation of PSGL-1 peptides by the GalNAc-Ts

To survey the role of flanking negative charge on the targeted glycosylation of Thr57 we characterized the apparent rates of glycosylation of the 5 PSGL-1 peptides in Fig. 7 against a series of GalNAc-Ts. We examined (see Fig. 8 and Supplementary Fig. S11) 5 negative charge substrate preferring transferases (GalNAc-T1, −T2, −T6, −T7 and -T11), one positive charged preferring transferase (GalNAc-T3) and 3 unique N and C- terminal charge preferring transferases (GalNAc-T4, −T5 & -T12) (Ballard et al. 2023). We anticipated that PSGL-1 specific transferases would be the GalNAc-Ts preferring negative charged substrates that would give the highest activity against the nR and nn (WT) (−4 and − 6 charge, respectively). PSGL-1 peptides. Surprisingly, we only found that GalNAc-T1 and -T2 showed this pattern (Fig. 8A and F) with GalNAc-T1 showing the highest rates towards the nn (WT) peptide compared to the other GalNAc-Ts examined. Interestingly, GalNAc-T6 showed the opposite, preferring the more positive PSGL-1 peptides (Fig. 8E) while GalNAc-T7, a glycopeptide preferring transferase, was relatively inactive showing greatest activity against the NN peptide (Fig. 8H). GalNAc-T11, which has the highest preference for negative charged substrates (Ballard et al. 2023) was one of the least active transferases although it showed highest preference towards the most negative nn (WT) peptide (Fig. 8I). GalNAc-T3, −T4 and -T5 gave preferences for the more positive substrates (Fig. 8C, 8B, and 8G) while GalNAc-T12 gave highest activity against the nR substrate, consistent with its specificity for N-terminal negative and C-terminal positive charges (Fig. 8D) (Ballard et al. 2023).[1] Overall, these data support the notion that GalNAc-T1 may be the primary GalNAc-T initiating glycosylation at PSGL-1 Thr57 and that this is dominated by the high negative charge surrounding this site.

Glycosylation of PSGL-1 Glycopeptides by C1GALT1

For the WT-*Dm*C1GALT1 and GFP-*Dm*C1GALT1, we observed (Fig. 9A and B) elevated preferences towards the most negatively charged PSGL-1 substrates (nR and nn (WT)),

[1] Interestingly, our ISOGlyP predictions of O-glycosylation (Mohl et al. 2020) give the highest enhancement value products (EVPs) for the mixed peptide/glycopeptide transferases GalNAc-T4 and -T12 (Revoredo et al. 2016) for glycosylating the PSGL-1 peptides (EVPs ~4–8) (Supplementary Figure S12) while the remaining transferases including GalNAc-T1 have much lower enhancement values (EVPs ~1–2), with GalNAc-T3 predicted to be unable to glycosylate the PSGL-1 peptides (EVP ~0.3). As the EV values are unique for each transferase their magnitude should not be compared between transferases, nevertheless, EVPs greater than 1.0 would indicate the ability to glycosylate the PSGL-1 sequence. It is noteworthy that, except for GalNAc-T12, the ISOGlyP +/−5 derived EVPs for the differently charged PSGL-1 peptides in Fig. 8 largely do not vary, while the case of GalNAc-T12, ISOGlyP incorrectly predicts the RR glycopeptide to be most active. The above discrepancies arise from the fact that ISOGlyP only examines +/− 5 residues from the acceptor and to the powerful role of remote flanking substrate charge on modulating GalNAc-T specificity.

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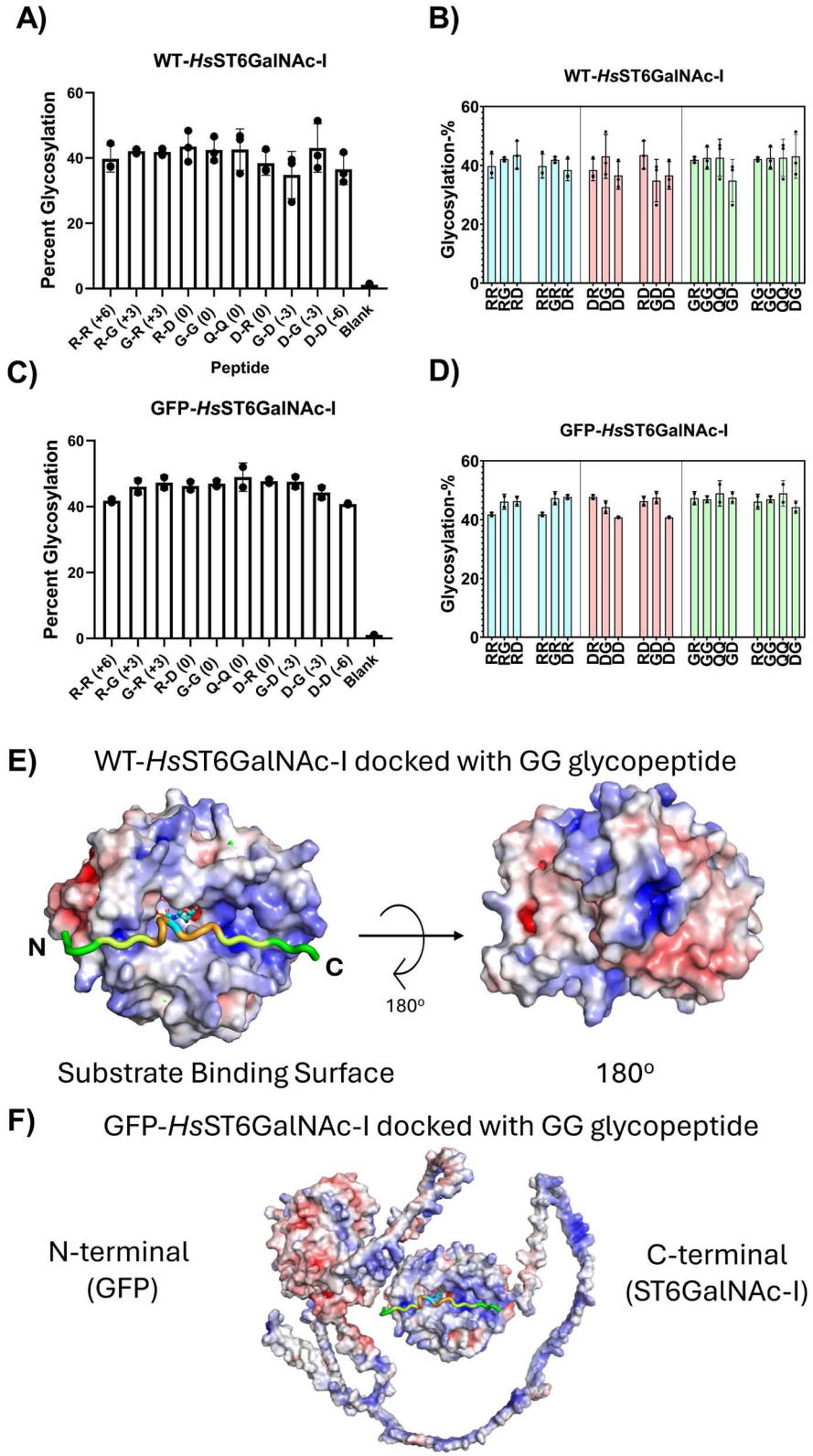
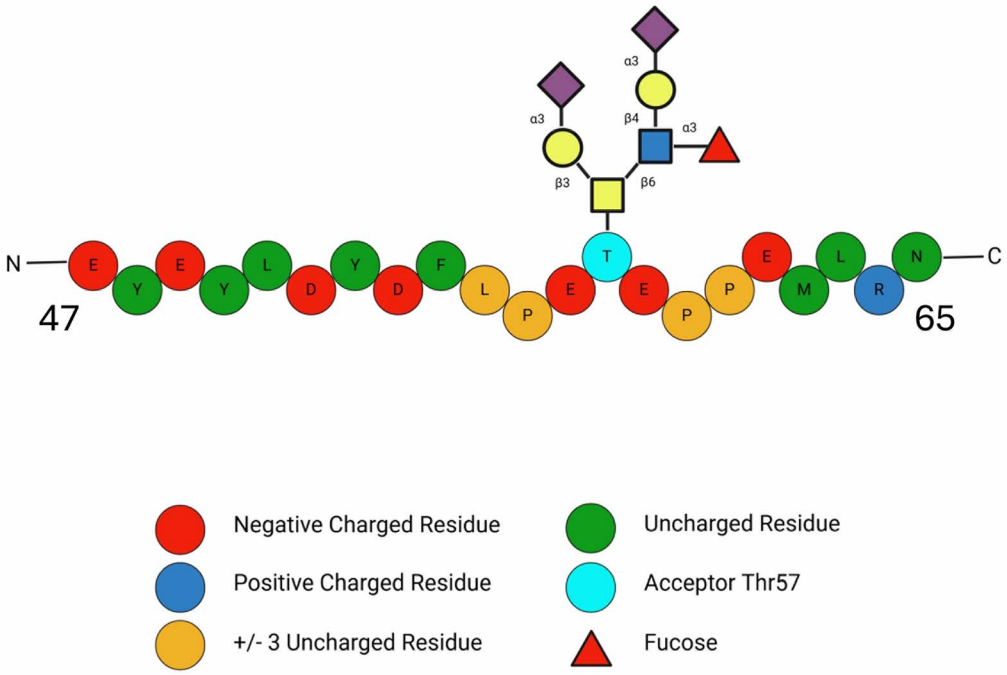


Fig. 6. *Hs*ST6GalNAc-I shows No specific charge preferences towards the semi-synthetic charged Glycopeptides in Table 1. Glycosylation of the charged glycopeptides (0.47 mM) in Table 1 by WT-*Hs*ST6GalNAc-I (0.02 μM) and GFP-*Hs*ST6GalNAc-I (0.02 μM), panels **A–D** (arranged as in Fig. 2A-D). **E** and **F**) Electrostatic surface charge of modeled WT- and GFP-*Hs*ST6GalNAc-I with the neutral GG glycopeptide substrate bound, color coded as in Fig. 2E and F, where the light green represents GAG neutral residues. See supplementary fig. 8 for statistical analysis. Experiments were repeated two to three times.

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A)



B)

PSGL-1 (Glyco)peptide Substrates

XYXYLXYXF-LPE(T/T*)EPP-XMLXN			X= Charged Residues of WT Sequence
Substrate ID [a]	Substrate Sequence [b]	Net charge [c]	N-/C- terminal charge [d]
RR	RYRYLRYRF-LPE(T/T*)EPP-RMLRN	+4	+4/+2
Rn	RYRYLRYRF-LPE(T/T*)EPP-EMLRN	+2	+4/0
NN	QYQYLQYQF-LPE(T/T*)EPP-QMLNN	-2	0/0
nR	EYEYLEYEF-LPE(T/T*)EPP-RMLRN	-4	-4/+2
WT (nn)	EYEYLEYEF-LPE(T/T*)EPP-EMLRN	-6	-4/0

Fig. 7. PSGL-1 Thr57 charged Glycopeptide substrates. **A**) Cartoon of the WT PSGL-1 Glu47-Asn65 glycopeptide with a sialyl Lewis-X structure attached to Thr57. Residue colors are defined in the fig. **B**) Table of differently charged PSGL-1 based peptide (T) and glycopeptide (T∗) substrates studied; named RR, Rn, NN, nR, and nn (WT) to represent positive charge (R for Arg) or neutral charge (N for Gln or Asn) residue N and C terminal substitution, where n represents the native N- or C-terminal sequence. The net peptide charge and flanking N- or C-terminal charges are given to the right of the sequences. Note the flanking sequence around Thr57, LETEPP, does not change in the different glycopeptides. (Fig. 7A was created in BioRender https://BioRender.com/l63u057.)

consistent with our findings against charged glycopeptides above (Fig. 2). However, the most negative nn (WT) glycopeptide was somewhat less active than the nR peptide containing C-terminal positive charges for WT-*Dm*C1GALT1 and even less so for the GFP version. An examination of the WT and GFP structures with the nR glycopeptide bound reveals a large negative patch in the GFP-*Dm*C1GALT1 which may account for these differences (Fig. 9C and D). This further suggests that both the position of flanking charged residues and overall charge of the substrates may have combined effects on the specificity of C1GALT1.

Glycosylation of PSGL-1 Glycopeptides by ST6GalNAc-II

For *Hs*ST6GalNAc-II, we observed a clear preference for the nR substrate showing a strong preference for an N-terminal negative and C-terminal positive charge (Fig. 10A), consistent with what we observed for WT-*Hs*ST6GalNAc-II (Fig. 5). The electrostatic surface of *Hs*ST6GalNAc-II shows a highly positive charged pocket near where we predict distant N-terminal charged residues may bind potentially enhancing the binding and sialylation of the nR substrate as shown in in the model of the glycopeptide bound in Fig. 10B. These results confirm

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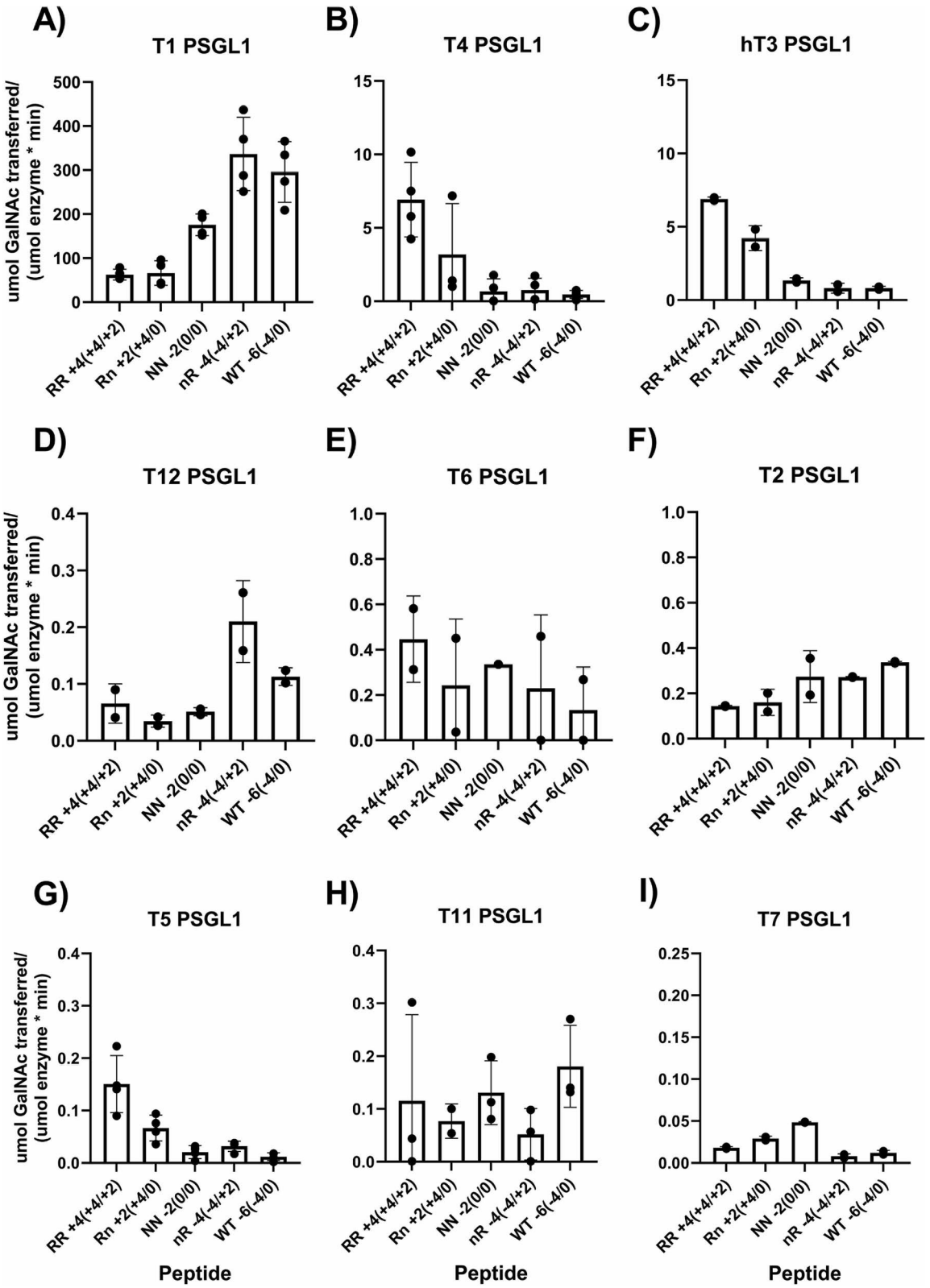


Fig. 8. Glycosylation of the differently charged PSGL-1 peptide substrates by GalNAc-Ts. Apparent rates of glycosylation of the PSGL-1 peptides (Fig. 7B) against a series of GalNAc-Ts where most positive peptides (RR) are on the left and the most negative at the right (WT (nn)). **A-I**) GalNAc-T data arranged in order of decreasing apparent rates to convey relative GalNAc-T activity. Experiments were repeated two to four times.

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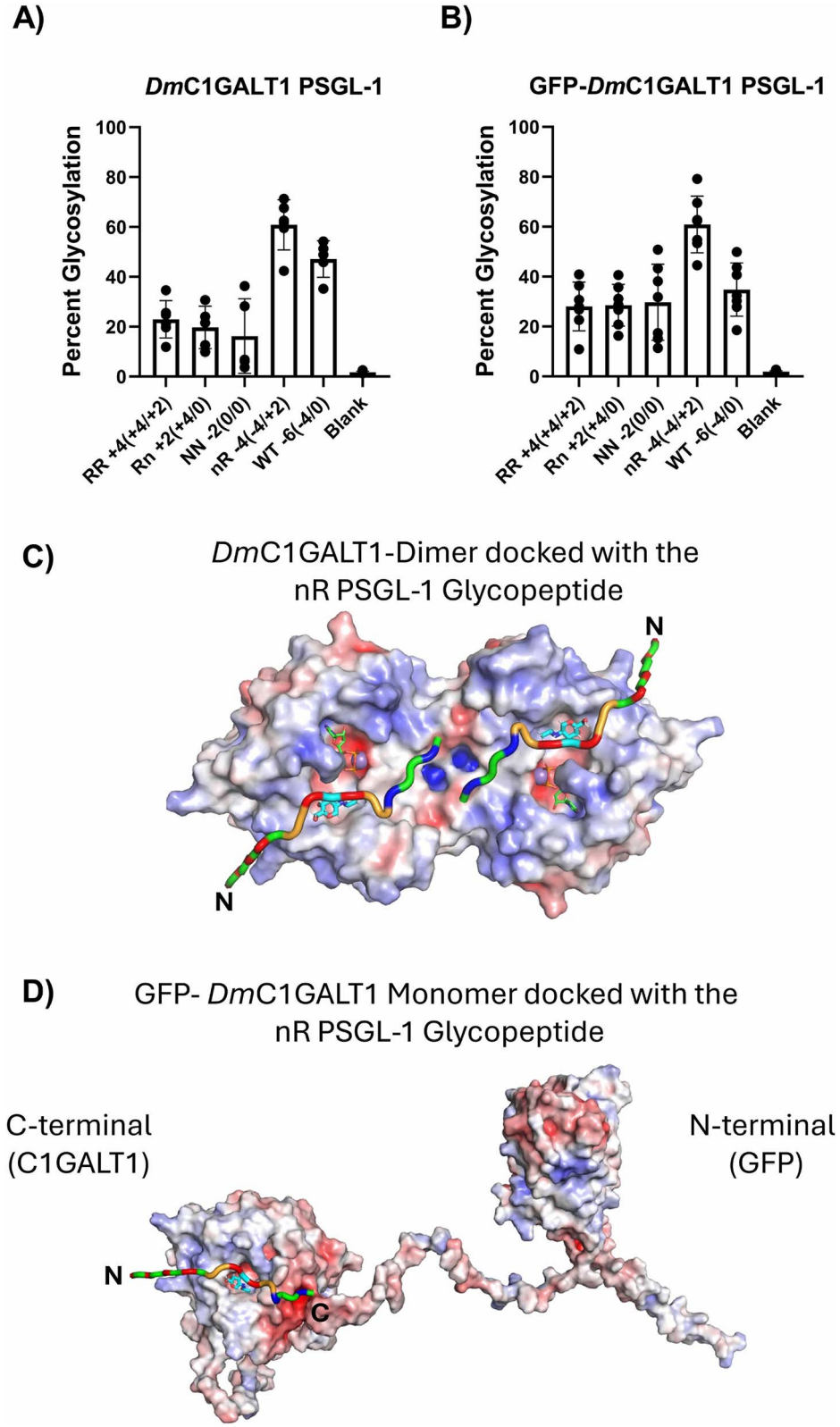


Fig. 9. WT-/GFP-*Dm*C1GALT1 shows negative charge preferences against PSGL-1 charged Glycopeptides. Glycosylation of the charged PSGL-1 glycopeptides (0.46 mM) in Fig. 7B against WT-*Dm*C1GALT1 (0.027 μM) and GFP-*Dm*C1GALT1 (0.027 μM) (panels **A** and **B** respectively). In **A**) and **B**) substrates are arranged with the most positive RR PSGL-1 glycopeptide on the left and the most negative WT (nn) PSGL-1 glycopeptide is on the right. **C** and **D**) the surface electrostatic charge of WT-*Dm*C1GALT1 and GFP-*Dm*C1GALT1 modeled with the nR PSGL-1 glycopeptide respectively. Glycopeptides are colored as described in Fig. 7. See Supplementary Fig. S2 for statistical analysis. Experiments were repeated five times.

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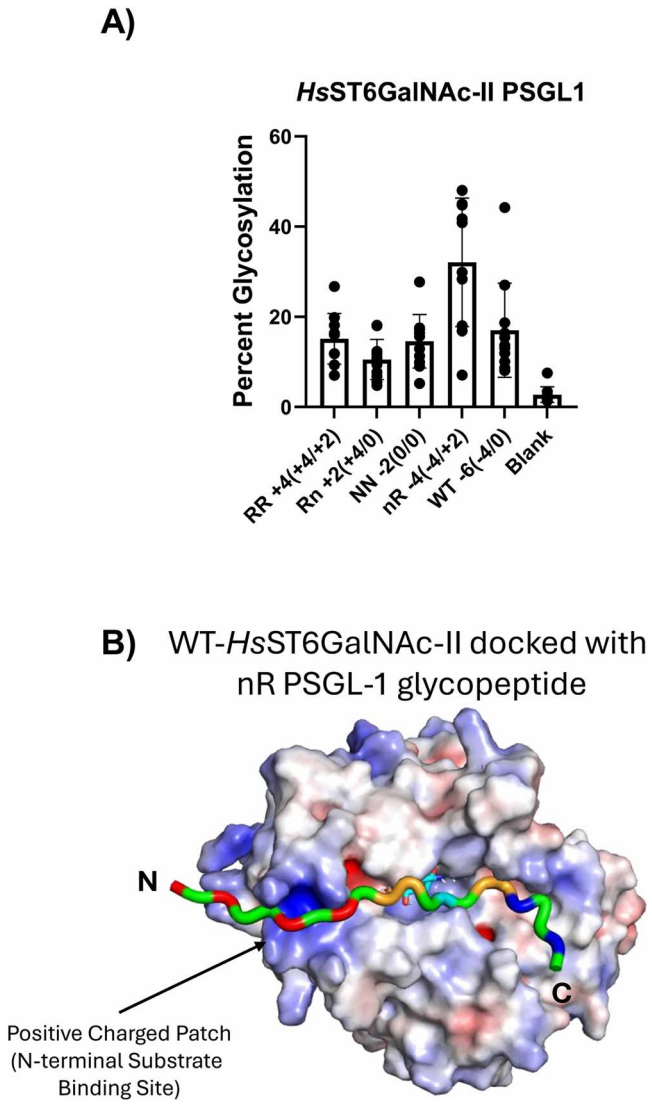


Fig. 10. WT-*Hs*ST6GalNAc-II reveals N-terminal negative and C-terminal positive substrate charge preferences against PSGL-1 substrates. Glycosylation of the charged PSGL-1 glycopeptides (0.46 mM) (Fig. 7B) by WT-*Hs*ST6GalNAc-II (0.02 μM) in **A**) and the calculated surface electrostatic distributions of WT-ST6GalNAc-II modeled with the nR PSGL-1 glycopeptide bound in **B**) Bound substrate is color coded as described in Fig. 7. See supplementary fig. 7 for statistical analysis. Experiments were repeated seven times.

that the pattern of charged residues, significantly impacts *Hs*ST6GalNAc-II specificity. Additionally, these results further suggest that *Hs*ST6GalNAc-II may preferentially target the T-antigen for sialylation as shown by its similar charge preferences with C1GALT1 against the PSGL-1 substrates, thus distinguishing it from *Hs*ST6GalNAc-I in these studies (see below).

Glycosylation of PSGL-1 Glycopeptides by ST6GalNAc-I

To our surprise, for *Hs*ST6GalNAc-I we observed a clear preference towards PSGL-1 glycopeptide substrates (RR and Rn) with N-terminal positive charge (Fig. 11A). This specificity was not observed in our charged glycopeptide substrates (Table 1 and Fig. 6). However, an examination of the surface charge of the *Hs*ST6GalNAc-I model, a distant pocket of negative charge is found in which remote N-terminal positive charged residues of a substrate could potentially bind thus enhancing glycosylation (Fig. 11B). To confirm the presence of this unique charge–charge interaction we performed experiments at high ionic strength (400 mM) designed to disrupt this interaction (Fig. 11C and Supplementary Fig. S10). At high ionic strength a dramatic decrease in activity towards the positive charged PSGL-1 glycopeptide substrates of ~5.5-fold was observed with minimal effects observed on the other substrates (Fig. 11D), thus confirming this specific charge–charge interaction in Fig. 11B**.** Collectively, these results show that *Hs*ST6GalNAc-I is indeed influenced by more remote substrate charge, potentially revealing a charge substrate motif for this transferase.

Discussion

Core O-glycan elongation is critical for embryonic development and important for regulating multiple biological pathways such as integrin signaling and angiogenesis (Ju et al. 2013; Akita et al. 2001; Ju et al. 2011; Yoshida et al. 2008; Xia et al. 2003; Kuo et al. 2021). Aberrant core glycosylation is linked to several disorders including many types of cancers, the Tn syndrome, ulcerative colitis, and inflammatory bowel disease (Ju et al. 2013; Munkley 2016; Bergstrom et al. 2017; Yao et al. 2022). Presently, it is not well understood how the core transferases select their glycoprotein targets. Having such information will contribute to identifying their specific targets and understanding their biological roles. In the present study, we examined the roles of glycopeptide substrate flanking charge on the core O-glycan elongating transferases (*Dm*C1GALT1, *Hs*B3GNT6, *Hs*ST6GalNAc-I, and -II) using a series of differently charged glycopeptides as well as a series of differently charged PSGL-1 Thr57 glycopeptides. Our results show that flanking substrate charge significantly impacts the specificity of *Dm*C1GALT1, preferring negatively charged substrates, while flanking charge variably impacts the specificities of *Hs*B3GNT6 and *Hs*ST6GalNAc-II each having unique N- and C-terminal substrate charge preferences. On the other hand, *Hs*ST6GalNAc-I appeared to have no significant charge preferences, but revealed a preference on the PSGL-1 substrates for remote N-terminal positive charged residues, thus uncovering a potential substrate motif for this transferase. Collectively, these studies identify another factor that variably impacts the specificities of the core elongating transferases.

These charge effects were shown to roughly correlate to the electrostatic surface potentials of the enzymes which was confirmed by disrupting interactions at high ionic strength. The negative charge preferring *Dm*C1GALT1 shows regions of high positive charge surrounding the known substrate binding site (Figs. 2 and Supplementary Fig. S2) while *Hs*B3GNT6 shows a complex pattern of charge preferences and dispersed regions of weak positive and negative charge surrounding its putative substrate binding site (Fig. 4 and Supplementary Fig. S6). *Hs*ST6GalNAc-II which shows unique N- and C-terminal charge preferences shows surface electrostatics consistent for binding N-terminal positive and C-terminal negative charged substrates (Fig. 5 and supplementary Fig. S7). In contrast, *Hs*ST6GalNAc-I which revealed no specific charge preference against our charged glycopeptide substrates shows dense regions of positive charge surrounding its predicted substrate binding

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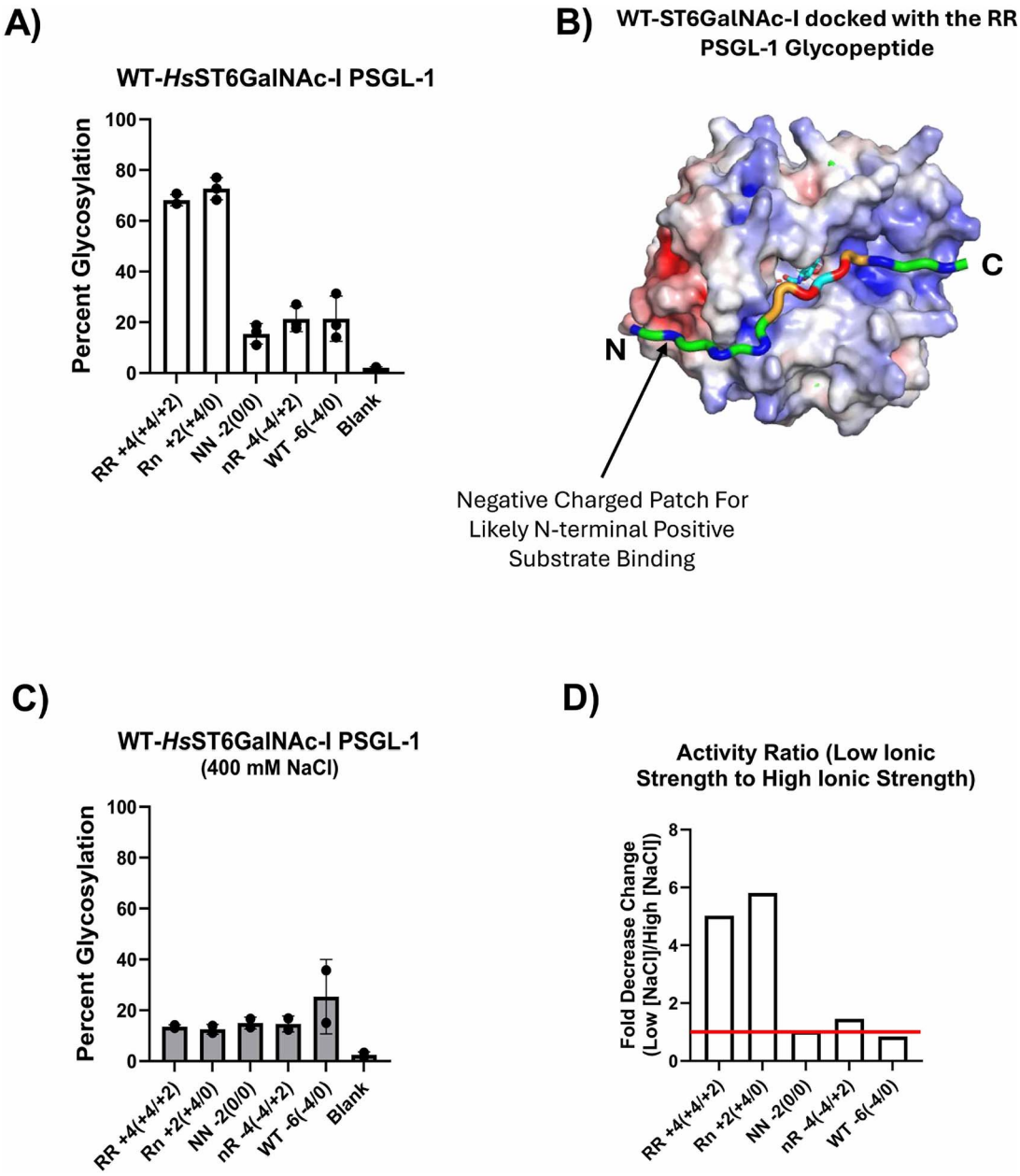


Fig. 11. WT-*Hs*ST6GalNAc-I reveals selectivity towards PSGL-1 substrates with remote N-terminal positive charged residues. Glycosylation of the charged PSGL-1 glycopeptides (0.46 mM) (Fig. 7B) by WT-*Hs*ST6GalNAc-II (0.02 μM) in **A**) and the electrostatic surface charge of *Hs*ST6GalNAc-I modeled with the RR PSGL-1 glycopeptide in **B**) (color coded in Fig. 7)**.** Note that the highlighted negative charged region where the remote N-terminal positive charged residues may interact. **C**) Effects of glycosylation at elevated (400 mM) ionic strength. **D**) Fold change of activity 400 mM vs 100 mM where the red line represents no change. Experiments were repeated two to three times.

site (Fig. 6 and Supplementary Fig. S8). This may suggest that *Hs*ST6GalNAc-I's specificity is mainly driven by the Tn antigen due to a higher accessibility of the GalNAc C6 hydroxyl group to this transferase rather than charge–charge effects. However, our findings of enhanced *Hs*ST6GalNAc-I activity against the N-terminal positive charged PSGL-1 are consistent with a distant patch of negative charge (Fig. 11 and Supplementary Fig. S8). This remote charge–charge interaction may serve to "anchor" specific substrates to the transferase. Altogether, our results indicate that substrate charge variably influences the substrate specificities of the core-O-glycan elongating transferases. Thus, these transferases likely utilize their unique patterns of surface charge, as well as other features such as peptide sequence/properties and density of neighboring glycosylation for selecting their glycopeptide substrates. However, due to the extended range of charge–charge interactions compared to shorter range hydrophobic or hydrophilic interactions, substrate charge will likely play a significant role, particularly for C1GALT1.

We have previously suggested that the targeted synthesis of the sialyl Lewis-X structure at Thr57 of PSGL-1 may be due to its high local negative charge (Gerken et al. 2011). We therefore characterized a series of differently charged PSGL-1 peptides and glycopeptides against a series of GalNAc-Ts and core elongating transferases. We found that the negative charge preferring GalNAc-T1 is likely the major

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GalNAc-T to glycosylate Thr57 in the native PSGL-1 and that the more positive substrates showed significantly lower activity (Fig. 8 and Supplementary Fig. S9). Similarly, *Dm*C1GALT1 also prefers the most negatively charged PSGL-1 glycopeptides (native and nR), consistent with the PSGL-1 substrate optimized for rapid glycosylation by *Hs*C1GALT1 (Fig. 9 and Supplementary Fig. S3). Additionally, *Hs*ST6GalNAc-II but not *Hs*ST6GalNAc-I shows elevated preference for the negative PSGL-1 substrates consistent with our charged glycopeptide studies. These results suggest that Thr57's negative flanking charge may help facilitate rapid initiation by GalNAc-T1 and subsequent elongation by *Hs*C1GALT1, thus making this site the most kinetically favorable for sialyl Lewis-X formation.

Podoplanin, a type 1 transmembrane sialomucin-like glycoprotein, is another known target for *Hs*C1GALT1, where its core-1 glycosylation by *Hs*C1GALT1 is essential for proper embryonic development and lymphatic function (Fu et al. 2008). Podoplanin has several possible O-glycosylation sites located at its N-terminal end, including several in a series of negatively charged so called PLAG repeats. These repeats have an EDXXXT/S sequence motif where the Thr or Ser contain a mono- or disialylated core-1 O-glycan and X is any amino acid residue (Kaneko et al. 2006). Thr34 and Thr52 of podoplanin's PLAG-1 and PLAG-3 repeats respectively, are crucial for platelet aggregation-inducing stability and are conserved and O-glycosylated across mammalian species (Kunita et al. 2007; Nagae et al. 2014). Thr52, specifically, contains a disialylated core 1 glycan that is necessary for interacting with the C-type lectin-like receptor-2 (CLEC-2) to promote platelet aggregation (Kaneko et al. 2007; Nagae et al. 2014). Interestingly, our sequence analysis of podoplanin reveals that Thr34 and Thr52 are located in the most negatively charged region on podoplanin (Fig. 12A). In addition, the PLAG repeats have been shown to be glycosylated by the negative charge preferring GalNAc-T1 and -T13 (Xu et al. 2016) thus targeting of these sites for subsequential glycosylation by *Hs*C1GALT1 similar to what we proposed for Thr57 of PSGL-1. Thus, podoplanin may be another example of *Hs*C1GALT1 using flanking negative charge to help target a biologically important substrate to ensure its biological function.

It has long been accepted that the GalNAc-Ts and core O-glycan elongating transferases will be broadly localized in the Golgi depending on their role in the pathway allowing for proper O-glycan initiation and elongation (Röttger et al. 1998; Brockhausen 2006; Bennett et al. 2012; Ju et al. 2014; Wandall et al. 2021; Harada et al. 2024; Sanji et al. 2024). The specificities we have observed for these transferases are consistent with such a hierarchy. We have shown that *Dm*C1GALT1's activity is highly modulated by flanking charge and that this charge effect is significantly greater than what we observed for the other core elongating transferases studied. Considering our results, and the range of specificities reported by others, it suggests that C1GALT1 is selective and should be localized primarily in the cis region of the Golgi to ensure proper core-1 glycosylation of its biologically significant sites i.e. PSGL-1, podoplanin, etc.[2] Once the C1GALT1 sites are glycosylated, the B3GNT6 and potentially ST6GalNAc-II would glycosylate subsequent unmodified Tn sites using their weaker charged motifs. Therefore, these transferases could be expected to be localized near the cis/medial Golgi. Finally, the low charge specificity of ST6GalNAc-I, observed in this study, would suggest that it would sialylate all remaining free GalNAc residues to block any remaining Tn antigen. This suggests that ST6GalNAc-I, which also has the highest preference towards Tn, should be localized in the late medial/trans Golgi to prevent the premature sialylation of biologically important C1GALT1 (and other transferase specific) sites. Our proposed hierarchy is consistent to that observed in cancer models, where the impaired expression of B3GNT6 or C1GALT1 disrupts core O-glycan elongation thus resulting in increased Tn and STn antigen expression. (Barrow et al. 2013; Munkley 2016; Ogawa et al. 2017; Wang et al. 2019; Luo et al. 2023). We acknowledge this is a very simplified model which hopefully will lead to further studies on the link between O-glycosyltransferase specificity and their localization in the Golgi. These findings are furthermore consistent with what is known about the Golgi localization of transferases and hydrolases involved in N-glycan biosynthesis where the enzymes are organized in the Golgi dependent on their hierarchy in N-glycan processing (Rabouille et al. 1995; Stanley et al. 2022; Yagi et al. 2024).

In summary, we have shown that the flanking charge of glycopeptide substrates is an important factor, among others, that influence the substrate specificities of the core O-glycan elongating transferases. Our results revealed *Dm*C1GALT1 to be the most sensitive and *Hs*ST6GalNAc-I the least sensitive to flanking charge. The preferences for *Dm*C1GALT1 are furthermore consistent with its targeted glycosylation of the biologically significant glycoproteins PSGL-1 and podoplanin. Our observed preferences also are consistent with the likely importance and localization of these transferases in the Golgi apparatus, where C1GALT1 would be early and ST6GalNAc-I late in the Golgi. Our studies are clearly one aspect of understanding how these transferases choose their target substrates based on their substrate specificity and Golgi localization. Therefore, further structural, biochemical, and cellular studies of these transferases are needed to further assess their specificity and to better understand their Golgi localization. In conclusion, we have shown that the core O-glycan elongating transferases have unique charge substrate specificities that aid in selecting their targeted glycosites; specifically advancing our understanding of how C1GALT1 may choose its biologically significant targets.

Methods

Reagents and (Glyco) peptide substrates:

Charge peptide substrates listed in Table 1 were custom synthesized by RS Synthesis (Louisville, KY). Stock solutions of 4 or 8 mM were prepared for each of the peptides after lyophilization from water multiple times and adjusting the pH to ~7 with 0.1 M NaOH/HCl. PSGL-1 based charged glycopeptides listed in Fig. 7 and the non-glycopeptide versions used in Fig. 8 were custom synthesized by Bachem Americas (Torrance, CA) and RS Synthesis (Louisville, KY), respectively. Stock solutions of 4 mM were prepared for each of the PSGL-1 glycopeptides after lyophilization and pH adjusted as described above. Concentrations of all peptides and glycopeptides were confirmed using a DeNovix DS-11 nanodrop spectrophotometer (DeNovix Inc., Wilmington,

[2] Interestingly, this would suggest the sites glycosylated by late acting, glycopeptide preferring GalNAc-Ts (such as GalNAc-T7 and -T10) would be less likely to be glycosylated by C1GALT1, due to its inhibition by prior glycosylation.

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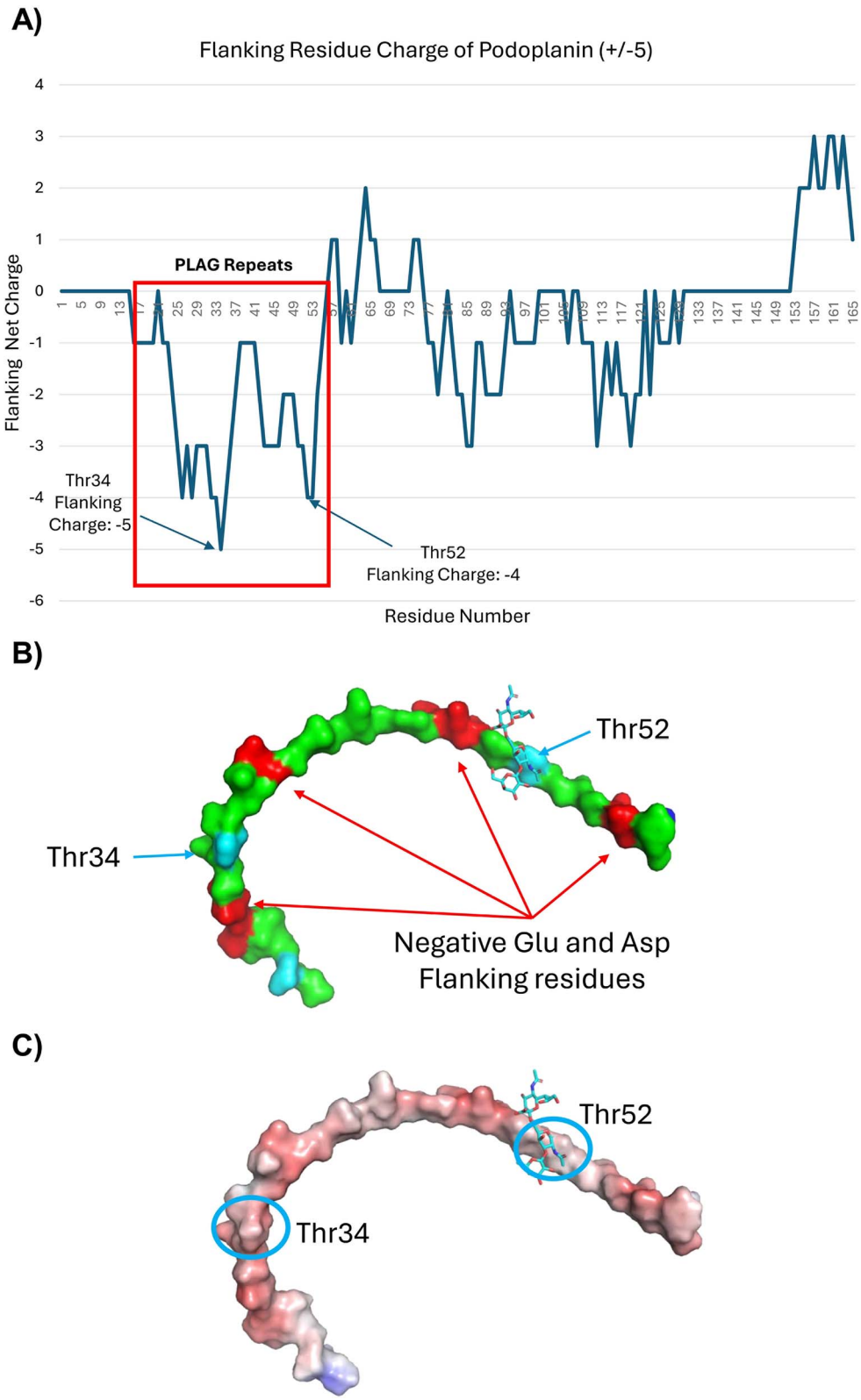


Fig. 12. Negative flanking charge may direct C1GALT1's glycosylation of Podoplanin's Thr34 and Thr52 in the PLAG 1 and 3 repeats. **A**) Plot of net flanking charge for *hs*Podoplanin showing highest net negative charge at Thr34 and Thr52**. (**plot obtained from the running +/− 5 residue sum of charge). **B**) Top: Model of the PLAG regions where Thr34 and Thr52 are highlighted in cyan, flanking negative charged residues are in red, and flanking uncharged residues are in green. **C**) Surface electrostatics of the PLAG regions. The disialylated core 1 glycan of Thr52 has been extracted from the crystal structure of PLAG-3 (PDB: 3WSR).

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DE) using the extinction coefficient of 1490 M − 1 cm − 1 (charged peptides) and 4470 M − 1 cm − 1 (PSGL-1 glycopeptides) at 280 nm based on the extinction coefficient of a peptide Tyr residue (Gasteiger et al. 2005). The sequence of all peptide and glycopeptides substrates were confirmed by Edman amino acid sequencing on a Shimadzu PPSQ 53A (Shimadzu Scientific Instruments Inc., Columbia, MD). Benzyl GalNAc was obtained from Millipore-Sigma (St. Louis, MO) where stock solutions of 8 mM pH ∼7 were made. Radiolabeled UDP-[^{3}H]GalNAc, UDP-[^{3}H]Gal, UDP-[^{3}H]GlcNAc, and CMP-[^{3}H]Neu5Ac were obtained from American Radiolabeled Chemicals Inc (St. Louis, MO), nonlabelled UDP-GalNAc, UDP-Gal, UDP-GlcNAc, and CMP-Neu5Ac were obtained from Millipore-Sigma (St. Louis, MO). Stock solutions of radiolabeled 20 mM UDP-GalNAc UDP-Gal, UDP-GlcNAc, and CMP-Neu5Ac were prepared by adding stock radiolabeled sugar donor to 20 mM unlabeled sugar donor giving ∼6 × 10^8 DPM/μmole. ScintiVerse BD Cocktail fluid was obtained from Fisher Scientific (Pittsburgh, PA). Liquid scintillation counting was performed on a Beckman LS 6500 Scintillation Counter. BioPureSPN TARGA C18 macro spin columns were obtained from The Nest Group Inc. (Ipswich, MA).

Transferases

Transferases were obtained as purified soluble N-terminal truncated enzymes from multiple sources and expression systems. Wild type *D. melanogaster* C1GALT1 was obtained from Ramon Hurtado-Guerrero (University of Zaragosa, SP) and expressed from High Five ™ cells as described in González-Ramírez et al. 2022. WT- and GFP-tagged versions of *Hs*ST6GalNAc-I and *Hs*ST6GalNAc-II as well as GFP-*Dm*C1GALT1 and GFP-*Hs*B3GNT6 were obtained from Kelley Moremen (CCRC, University of Georgia) and expressed from HEK293F cells (Moremen et al. 2018). The sequences of the expressed core transferases used in this study are given in Supplementary Fig. S14. The GalNAc-Ts used in this study were as described in (Ballard et al. 2023).

Intrinsic glycosylation of recombinant transferases

To investigate whether any potential in-vivo glycosylation may impact the charge specificity of C1GALT1, B3GNT6, ST6GalNAc-I, and ST6GalNAc-II transferases we utilized the Glygen database (GlyGenHomeInformaticsResourcesforGlycoscience|glygen.org) to identify and map the known and predicted sites of N-glycosylation and O-glycosylation (Supplementary Fig. S13). The GFP- and WT-*Dm*C1GALT1 utilized in study (expressed from different expression systems) have no N-glycan sequons and are not expected to be O-glycosylated (González-Ramírez et al. 2022). We found in the remaining transferases, the location of glycosylation sites were distant from the predicted glycopeptide binding regions. This suggests that any N- and O-linked glycosites in the B3GNT6, ST6GalNAc-I and -II, when recombinantly expressed in-vivo, will likely possess similar charge preferences as determined in this study.

The semi-synthetic approach (control studies)

The charged glycopeptide substrates (Table 1) used to evaluate the role of flanking charge on the core elongating transferases were obtained by glycosylating the charged peptide substrates previously reported in Ballard et al. 2023 by GalNAc-T1. One-hundred percent GalNAc glycosylation was achieved by overnight incubations with GalNAc-T1 (1.4 μM) and 2 mM UDP-[^{3}H]GalNAc (abbreviated in Figures as UDP-[^{3}H]GaN) and verified via C18 spin column chromatography and scintillation counting (Supplementary Fig. S1A). As shown in Supplementary Fig. S1B and C, the C1GALT1 transferase (0.027 μM) does not utilize UDP-[^{3}H]GalNAc (2 mM) to glycosylate the "naked" or preformed glycopeptide substrate (0.7 mM), respectively. Thus, excess UDP-GalNAc should not impact the core transferase's ability to glycosylate the charged glycopeptide substrates. On the other hand, we found that GalNAc-T1 could utilize UDP-[^{3}H]Gal (2 mM) to glycosylate the "naked" charged peptide substrates while showing preferences towards the negatively charged substrates, similar to using UDP-[^{3}H]GalNAc as reported in Ballard et al. 2023. On the other hand, GalNAc-T1 does not utilizing UDP-[^{3}H]Gal (2 mM) to glycosylate synthetic GalNAc glycopeptide substrates (0.7 mM) (obtained from Sussex Research, Ottawa, CN) (Supplementary Fig. S1D and E). (Note that these glycopeptide substrates were not utilized in the remainder of the study due to our interevent introduction of contaminating species that selectively inhibits C1GALT1 transferase activity, manuscript in preparation.

The semi-synthetic approach (core transferase activity assays)

Semi-synthetic glycopeptides were produced from "naked" peptide substrate (0.7 mM) in sodium cacodylate buffer (100 mM pH 6.8) containing $MnCl_2$ (10 mM), cold (non-radiolabeled) UDP-GalNAc (2 mM), and GalNAc-T1 (1.4 μM) in a final volume of 20 μL. An overnight incubation at 37 °C gave 100% glycosylation (Supplementary Fig. S1A). Next, 10 μL of an additional master mix was added containing sodium cacodylate buffer (33 mM pH 6.8), $MnCl_2$ (10 mM) for C1GALT1 and B3GNT6, appropriate sugar donor (2 mM), and core glycosyltransferase (0.027–1.2 μM) giving a final concentration of glycopeptide substrate of 0.47 mM. The reaction mix was incubated at 37 °C for 1–3 hrs. Workup followed that previously described in (Ballard et al. 2023) with minor differences. After incubation, reactions were quenched with 200 μL of 0.5% TFA in H_2O. Ten percent (23 μL) of the reaction volume was removed for [^{3}H] scintillation counting (initial DPM), and the remainder was applied to an equilibrated TARGA C18 hydrophobic spin columns and spun for 1 minute and collected. A wash with 800 μL of 0.1% TFA in H_2O eluted, depending on the transferase, free UDP-[^{3}H]Gal, [^{3}H]Gal, UDP-[^{3}H]GlcNAc, [^{3}H]GlcNAc, CMP-[^{3}H]Neu5Ac, or [^{3}H]Neu5Ac. The bound ^{3}H glycopeptide and unmodified GalNAc glycopeptides were eluted using two washes of 200 μL of 50/50 acetonitrile in 0.1% TFA followed by 200 μL of 100% acetonitrile, each spun for 1 min at 800 rpm, and a final 100 μL of 100% acetonitrile spun for 4 min. The extent of ^{3}H glycopeptide glycosylation was calculated utilizing the total ^{3}H and bound ^{3}H DPM as described in (Ballard et al. 2023). Reactions were typically repeated three or more times as shown in the plots. GraphPad Prism version 9.2 for Windows (GraphPad Software, San Diego, CA) was used to plot the data and perform statistical analysis. Note for B3GNT6 reactions the data in Fig. 3 were normalized between experiments to take account for decreases in specific activity over time.

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PSGL-1 GalNAc-T studies

Final reactions consisted of sodium cacodylate buffer, pH 6.8, 1 mM β-mercaptoethanol, 0.1% Triton X-100, UDP-[^{3}H]GalNAc (0.2–2 mM), enzyme (0.006–1.17 μM), peptide substrate (0.35–1.4 mM), and were incubated at 37 °C for 20 min to overnight depending on GalNAc-T activity. After incubation, reactions were quenched and worked up as described above and in (Ballard et al. 2023). The extent of ^{3}H glycopeptide glycosylation was calculated utilizing the total ^{3}H and bound ^{3}H DPM as described in (Ballard et al. 2023). Percent peptide glycosylation values were converted to μM of GalNAc transferred/(μM enzyme $*$ min) using the initial amounts of UDP-GalNAc, substrate, and enzyme present in the reaction. Reactions were typically repeated two or more times as shown in the plots. GraphPad Prism version 9.2 for Windows (GraphPad Software, San Diego, CA) was used to plot the data and perform statistical analysis.

PSGL-1 Glycopeptide studies

Final reactions consisted of PSGL-1 glycopeptides (0.46 mM), sodium cacodylate buffer (100 mM pH 6.8) containing $MnCl_2$ (10 mM), appropriate sugar donor (2 mM), and core glycosyltransferase (0.003–0.027 μM) in a final volume of 20 μL. After incubation, reactions were quenched and worked up as described above. The extent of ^{3}H glycopeptide glycosylation was calculated utilizing the total ^{3}H and bound ^{3}H DPM as described in (Ballard et al. 2023). Reactions were typically repeated three or more times as shown in the plots. GraphPad Prism version 9.2 for Windows (GraphPad Software, San Diego, CA) was used to plot the data and perform statistical analysis.

Protein modeling

AlphaFold 3 was used to obtain WT structures of *Hs*B3GNT6 and *Hs*ST6GalNAc-I as well as the GFP tagged versions of *Dm*C1GALT1, *Hs*B3GNT6, *Hs*ST6GalNAc-I, and -II. Input enzyme sequences for AlphaFold 3 modeling corresponded to the sequences of the expressed and purified transferases (Supplementary Fig. S14). Dimeric *Dm*C1GALT1 (7Q4I) and *Hs*ST6GalNAc-II (6APL) structures were obtained from the Protein Data Bank. AutoDock Vina was used to predict potential GalNAc glycopeptide binding sites utilizing the truncated -YAVT$*$PGP- sequences where T$*$ represents GalNAc-O-Thr. PyMOL was utilized to manipulate GFP linkages to provide a clearer exposure to predicted glycopeptide binding sites, for alignment of charged glycopeptide and PSGL-1 charged glycopeptide substrates, and for calculation of surface electrostatics by the APBS extension following the default parameters (see Supplementary Fig. S4 **for scaling key**). The 3-button mouse editing tool was used to orient the charged regions of the glycopeptides into the proposed pockets of electrostatic interactions providing additional visuals of the suggested substrate-enzyme electrostatic interactions. Human C1GALT1 was modeled in AlphaFold 3 for the amino acid residues corresponding to the expressed *Dm*C1GALT1 (Supplementary Fig. S3).

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Journal online.

Authors' contributions

CJB: methodology, investigation, visualization, validation, supervision, software, writing-original draft, writing-reviewing and editing,
MRS: methodology, investigation, visualization, validation, writing-reviewing and editing,
LDC: methodology, investigation, visualization, validation, writing-reviewing and editing,
CKW: methodology, investigation, visualization, validation, writing-reviewing and editing,
HMA: methodology, investigation, visualization, validation, writing-reviewing and editing,
HKL: methodology, investigation, visualization, validation, writing-reviewing and editing,
DGC: resources, writing-reviewing and editing,
RH-G: resources, writing-reviewing and editing,
KWM: resources, writing-reviewing and editing,
TAG: conceptualization, project administration, funding acquisition, supervision, investigation, methodology, visualization, writing-original draft preparation, writing-review and editing. All authors approved the final manuscript.

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Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

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Conflict of interest statement

The authors declare no conflicts of interest.

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Data availability statement

The data underlying this article are available in the article and in its online supplementary materials. Any additional data not included will be shared on reasonable request to the corresponding author.

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